SIMATS ENGINEERING
Department of Information Technology
ITA05 – Computer Vision
ASSIGNMENT 1 – CO1

Assignment Title: Apply, Compare, and Justify Image Acquisition, Formation, and Processing Techniques for a Vision-Based E-Waste Battery Sorting Line

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.

Student Name	STHUTI SHARMA S	**Register Number**	192572072
Department / Section	Information Technology	**Academic Year / Semester**	2025–2026 / _____
Faculty Name	DR. MURALI P	**Nature of Assignment**	Individual
Duration	1 Week	**Total Marks**	50
Date of Submission	30.10.2026	**Date of Evaluation**	07.11.2026

Scenario – Vision-Based E-Waste Battery Sorting Line

GreenCycle Robotics Corp. operates an e-waste reclamation facility. They manually sort mixed consumer batteries coming off a bulk dump belt. The human operators experience rapid eye fatigue, miss dangerous structural swellings, and fail to read tiny laser-etched chemistry strings under changing factory ambient lights. Two sorting batches were recently rejected by the chemical smelter because lithium cells were mixed into a batch of nickel-metal hydride (NiMH) batteries, presenting an explosion hazard.

The facility requires a high-reliability computer vision system placed over an automated conveyor line to:

1. Classify form factors (Cylindrical 18650, Cylindrical 21700, and Prismatic pouch cells) for mechanical handling guides.
2. Read alphanumeric chemistry codes (e.g., "LCO", "LFP", "NMC", "NiMH") laser-etched onto glossy, reflective metallic casings.
3. Detect safety defects (surface rust spots, severe dents, and pouch-cell ballooning/swelling), then trigger a pneumatic piston kick-rejecter.

Before any deep-learning model is trained, the plant requires an analytically proven imaging front-end: the correct sensor, resolution, quantization parameters, optics, and lighting topology, organized into transparent processing levels.

As the Vision Engineer, you are given three candidate techniques in every task. You must apply all three mathematically or algorithmically, compare them, choose one, and justify your choice with concrete engineering evidence.

Operating Line Constraints & Parameters

- **Conveyor Belt:** Single lane, matte black rubber belt; speed = 1.2 m/s. Imaged Field of View (FOV) = 320 mm across the belt × 240 mm along the belt.
- **Target Objects:** Batteries ranging from 10 mm to 70 mm in length, up to 30 mm in height. Surfaces vary from polished, waxed glossy steel jackets to highly reflective plastic foil wrappers.
- **Line Throughput:** Up to 15 cells per second. The physical sorting decision must be committed within 400 milliseconds (the rejection piston is exactly 0.48 m downstream from the camera center).
- **Quality & Sensitivity Limits:** Smallest defect to detect is a 0.3 mm micro-crack/rust pit. Laser-etched text stroke-width is 0.4 mm. Pouch swelling must be caught with a dimensional precision of ±0.5 mm. Dark rust spot contrast is low, sitting at only 5–8 grey levels (on an 8-bit scale) away from clean brushed metallic casing.
- **Factory Lighting:** Open industrial shed with corrugated transparent plastic roofing sheets. Ambient light changes continuously between 1,500 lux (cloudy morning) and 45,000 lux (bright afternoon). Overhead tube lights run on a 50 Hz power grid, introducing 100 Hz illumination flicker.
- **Camera Geometry:** Fixed vertical overhead mount, exactly 400 mm above the conveyor belt plane looking straight down.
- **Compute Constraints:** Edge Industrial PC with 4 CPU cores, 16 GB RAM, no GPU. Native code stack must run using Python, OpenCV, and NumPy. Local archive storage capacity is limited to a 1.5 TB SSD.
- **Data Policy & Budget:** The entire hardware configuration budget is capped at ₹1,25,000. For compliance audits, the facility must store visual proof of every processed cell for 5 days. The factory's service contract mandates an absolute maximum misclassification/defect miss rate of ≤ 1.5%.

1. Executive Summary

The proposed system is a fixed, overhead machine-vision station that inspects each battery while it travels on a matte-black conveyor. Its purpose is to identify the battery form factor, read the chemistry marking, and detect visible safety defects before a downstream pneumatic reject point. The design must be deterministic, auditable, and achievable with Python, OpenCV, NumPy, four CPU cores, 16 GB RAM, and a total hardware ceiling of ₹1,25,000.

Figure: End-to-end vision system architecture



In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

2. Problem Definition and Industrial Context

Manual inspection is vulnerable to fatigue, inconsistent judgement, glare, small engraved characters, and changing ambient illumination. The assignment scenario describes rejected batches caused by lithium cells entering a nickel-metal-hydride stream. The engineering problem is therefore not merely image classification: it is a safety-critical inspection and traceability problem in which missed defects or chemistry errors can have serious downstream consequences.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

3. System Requirements and Acceptance Criteria

The system must classify cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells; read codes such as LCO, LFP, NMC, and NiMH; and detect rust, severe dents, and pouch swelling. The smallest stated defect is 0.3 mm, etched stroke width is 0.4 mm, swelling measurement tolerance is ±0.5 mm, and the service contract sets a maximum miss or misclassification rate of 1.5%.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

4. Operating Constraints and Design Implications

The belt moves at 1.2 m/s and the imaged field is 320 mm by 240 mm. Throughput reaches 15 cells per second, so the average cell interval can be as short as 66.7 ms. The piston is 0.48 m downstream from the camera and the decision deadline is 400 ms. These values drive the exposure time, image transfer, processing budget, and trigger scheduling.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

5. Functional Architecture

A robust architecture separates image acquisition, calibration, preprocessing, object localisation, feature extraction, classification, optical character recognition, defect inspection, decision fusion, and actuator control. Each stage should expose measurable inputs and outputs. This modular structure makes it possible to validate the imaging front-end independently before any deep-learning model is trained.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

6. End-to-End Process Flow

A photoelectric or encoder-based trigger identifies the inspection window. The camera captures a short-exposure image under controlled illumination. The processor corrects lens distortion and shading, segments the battery from the belt, identifies its region of interest, and performs parallel form-factor, text, and defect analyses. The results are fused into a traceable pass, reject, or uncertain decision.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

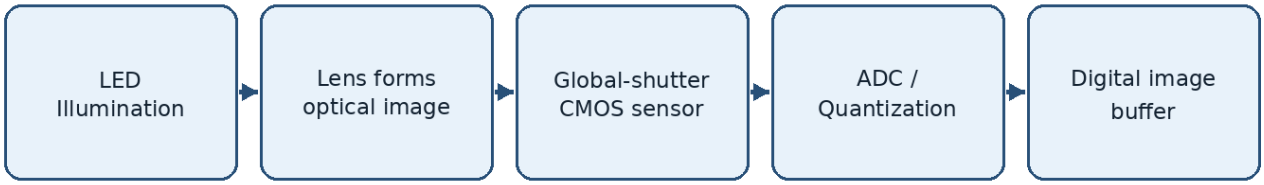
For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

7. Image Sensing Fundamentals

A digital camera converts incident optical energy into electrical charge at an array of photosites. The lens forms an image of the battery surface on the sensor, each pixel integrates light during exposure, and readout electronics convert the accumulated signal into digital values. Sensor size, pixel pitch, quantum efficiency, read noise, full-well capacity, and bit depth influence the final image quality.

Figure: Image acquisition and digitisation



In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

8. Image Formation Model

A simplified image-formation model is $I(x,y) = Q\{g\cdot[E(x,y)\rho(x,y)] + n(x,y)\}$, where E is illumination, ρ is surface reflectance, g represents sensor gain and optical throughput, n represents noise, and Q represents quantization. Glossy metal and foil produce strong specular components, so the observed intensity is not a direct measure of surface colour or material condition.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

9. Reflectance and Surface Appearance

Battery surfaces include brushed metal, polished steel, printed labels, and reflective polymer foil. Diffuse reflection spreads light over many directions, while specular reflection concentrates it around the mirror direction. A single uncontrolled overhead lamp can create saturated highlights that erase etched strokes or hide shallow dents. The optical design must therefore control the direction, size, and polarisation of illumination.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

10. Sampling and Spatial Resolution

Sampling converts a continuous optical image into a discrete pixel grid. The field width is 320 mm; if the camera provides 2448 horizontal pixels, the nominal scale is 7.65 pixels/mm. A 0.3 mm feature then spans about 2.29 pixels, which is marginal for reliable detection. A 4096-pixel width gives 12.8 pixels/mm and about 3.84 pixels across the same feature, providing more useful sampling margin.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

11. Nyquist Criterion and Feature Visibility

The Nyquist criterion states that a periodic signal needs at least two samples per cycle, but practical defect inspection needs more than the theoretical minimum. A micro-crack has irregular geometry and may have low contrast, so several pixels across its width are preferable. Optical blur, demosaicing, motion blur, and compression reduce effective resolution; the design should target at least three to five pixels across the smallest critical feature where feasible.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

12. Quantization and Bit Depth

Quantization maps an analogue sensor voltage into a finite number of digital levels. An 8-bit image contains 256 levels, while a 12-bit image contains 4096 levels. For a low-contrast rust spot only 5–8 grey levels away from clean metal, higher bit depth can preserve subtle intensity differences during acquisition and early processing. Processing should avoid unnecessary conversion to 8-bit until the relevant contrast operations are complete.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

13. Sensor Technology Comparison

A global-shutter CMOS sensor exposes all pixels at effectively the same instant, which is valuable for moving objects. A rolling-shutter sensor reads rows sequentially and can geometrically distort a fast-moving battery. CCD sensors can provide good image quality but are generally less attractive for a cost-constrained, high-frame-rate industrial system. A global-shutter industrial CMOS camera is the strongest practical candidate.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

14. Monochrome Versus Colour Sensor

A monochrome sensor dedicates each photosite to measuring intensity and typically offers higher spatial detail and sensitivity than a colour sensor of similar resolution. Colour can help distinguish printed markings or corrosion hues, but the task's most demanding requirements are geometry, engraved text contrast, and small defects. A monochrome camera with controlled white or multi-angle lighting is a strong baseline; a colour camera is justified only if colour features add validated discrimination.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

15. Camera Resolution Selection

The resolution should be selected from the smallest object feature, not from the overall battery dimensions. With a 320 mm field width, 4096 horizontal pixels provide 12.8 pixels/mm. At that scale, a 0.3 mm feature spans approximately 3.84 pixels and a 0.4 mm etched stroke spans approximately 5.12 pixels. This is a more defensible starting point than a low-resolution camera, although optical MTF and actual contrast must still be tested.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

16. Pixel Scale and Field-of-View Calculation

For a sensor width of N pixels covering 320 mm, the scale is N/320 pixels per millimetre. At N = 4096, the scale is 12.8 px/mm. The vertical scale for 3072 pixels over 240 mm is also 12.8 px/mm. A 70 mm battery would occupy about 896 pixels along its length when aligned with the corresponding image axis, leaving adequate pixels for shape measurement and region-of-interest processing.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

17. Lens Selection and Working Distance

The camera is fixed 400 mm above the conveyor. Lens focal length must provide the required field of view while keeping distortion, vignetting, and focus variation within calibration limits. A low-distortion industrial lens with a suitable image circle and aperture is preferred. The lens should be locked mechanically, and focus should be checked across the full belt plane rather than only at the centre.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

18. Depth of Field and Aperture

Battery height can reach 30 mm, so the object surface may lie above the belt plane. The lens aperture must provide sufficient depth of field to keep the top surface acceptably sharp at the minimum and maximum object heights. Stopping down increases depth of field but reduces light; stronger strobed illumination can compensate while retaining a short exposure.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

19. Lens Distortion Calibration

Radial distortion bends straight lines and can bias diameter, length, and swelling measurements. Tangential distortion arises from lens and sensor alignment. Calibration with a checkerboard or dot target at the belt plane estimates intrinsic parameters and distortion coefficients. The corrected image should be checked against known dimensions at the centre and corners of the 320 mm by 240 mm field.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

20. Geometric Calibration and Pixel-to-Millimetre Mapping

A planar calibration target placed on the conveyor establishes the mapping from image coordinates to physical coordinates. For a camera looking vertically down, a homography can map the belt plane to a metric reference plane. Since battery tops can be elevated, parallax and perspective effects should be measured; a fixed height reference or calibrated dimensional model can reduce errors in pouch swelling measurements.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

21. Exposure Time and Motion Blur

At 1.2 m/s, the object moves 1.2 mm in one millisecond. To limit motion blur to 0.1 mm, exposure should be no longer than 0.083 ms, approximately 1/12,000 s. This short exposure requires high-intensity pulsed lighting and a camera with suitable sensitivity. Encoder-triggered strobing can freeze motion more consistently than relying on continuous ambient light.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

22. Frame Rate and Throughput

A maximum of 15 cells per second corresponds to one cell every 66.7 ms on average. The camera and acquisition interface must sustain the required frame rate with margin for trigger jitter, image transfer, and occasional retries. A practical target is to complete acquisition and processing in substantially less than the 400 ms actuator deadline, while ensuring that consecutive objects do not overwrite or confuse processing buffers.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

23. Triggering and Conveyor Synchronisation

A photoelectric sensor can detect an approaching battery, while a conveyor encoder provides position and speed. Encoder position is preferable for scheduling the reject event because belt speed may vary. Each captured object should receive a unique sequence number and timestamp. The decision record is associated with the encoder position so the correct battery is rejected even when spacing is irregular.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

24. Lighting Challenges in the Factory

Ambient illumination varies from 1,500 lux to 45,000 lux and overhead lamps introduce 100 Hz flicker. Depending on ambient light would make image intensity unstable. The system should use a light-shielded inspection hood, high-intensity controlled illumination, short exposure, and synchronised strobes. Ambient light sensors can be used for diagnostics, but should not determine the main image brightness.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

25. Bright-Field Lighting

Bright-field illumination directs light toward the object and collects a significant reflected component. It can provide high signal levels and reveal printed labels or broad surface features. On polished metal, however, direct bright-field illumination can create glare and saturate pixels. It is useful when paired with diffusion, cross-polarisation, or a geometry that directs specular reflection away from the lens.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

26. Dark-Field Lighting

Dark-field illumination places the light at a low angle so smooth surfaces reflect little light into the camera while edges, scratches, pits, and raised defects scatter light toward the lens. This can make micro-cracks and rust pits more visible. Its limitation is sensitivity to object orientation and surface texture; multiple light directions or a ring arrangement can reduce directional blind spots.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

27. Diffuse Dome and Coaxial Lighting

A diffuse dome creates broad, even illumination and reduces hard reflections on curved or irregular surfaces. It is helpful for reading markings on shiny casings, although it can reduce the contrast of shallow topographic defects. Coaxial illumination is effective on flat reflective surfaces and can reveal certain printed or etched features, but curved cylindrical batteries may produce nonuniform response.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

28. Polarised Lighting Strategy

A polariser on the light and a crossed analyser on the lens can suppress specular reflection from non-metallic glossy surfaces and some coated materials. Metal surfaces may depolarise or preserve polarisation differently depending on finish and angle, so the benefit must be experimentally verified. Polarisation should be treated as one part of a multi-light strategy rather than a universal glare solution.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

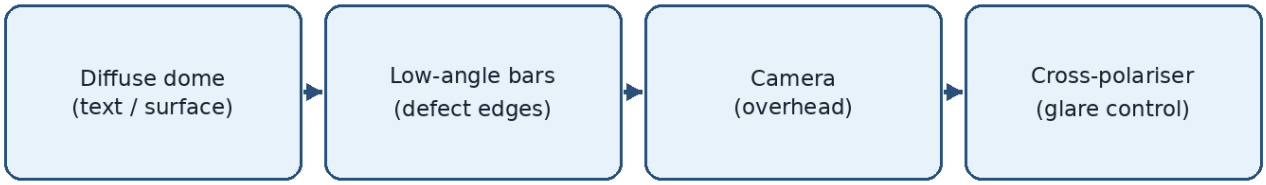
For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

29. Recommended Lighting Topology

A practical arrangement combines a diffuse high-angle source for general appearance, low-angle dark-field bars from multiple directions for scratches and pits, and a controlled coaxial or dome component for text. The lights should be independently switchable and synchronised to acquisition. A short sequence of exposures may improve robustness, but it must fit the throughput and compute budget.

Figure: Suggested multi-mode lighting arrangement



In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

30. Image Acquisition Hardware Configuration

The front-end should include a global-shutter monochrome industrial camera, a low-distortion lens, rigid overhead mount, strobed LED lighting, light enclosure, encoder, trigger sensor, and industrial PC. The selected camera should support hardware triggering and stable exposure control. The lens and camera must be protected from vibration, dust, and accidental adjustment.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

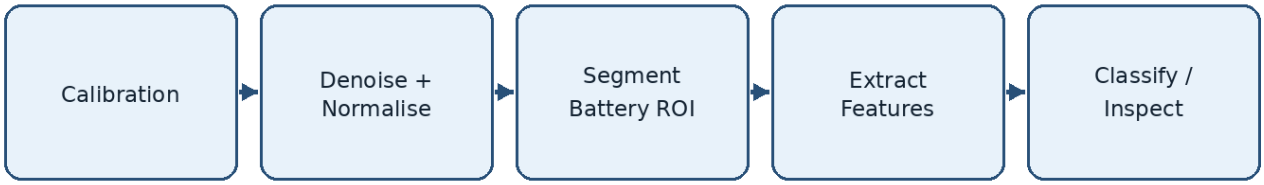
For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

31. Image Preprocessing Pipeline

Preprocessing prepares raw frames for reliable measurement. Recommended operations include dark-frame or flat-field correction where appropriate, lens undistortion, background estimation, denoising, contrast normalisation, and region-of-interest extraction. Each operation must be validated because excessive smoothing can erase 0.3 mm defects and aggressive contrast enhancement can amplify noise.

Figure: Image processing pipeline



In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

32. Noise Sources and Noise Reduction

Image noise can arise from photon statistics, read electronics, fixed-pattern response, thermal effects, and interference from lighting. Median filtering can suppress isolated impulse noise while preserving some edges; Gaussian filtering reduces high-frequency noise but blurs fine defects. A small kernel and conservative parameters are essential. Temporal averaging is generally unsuitable for moving objects unless frames are position-registered.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

33. Background Subtraction and Belt Segmentation

The matte-black belt provides a useful background contrast for many batteries. A reference background image can be maintained when the belt is empty, and foreground pixels can be detected using intensity difference or adaptive thresholding. Belt wear, dust, and shadows can cause false regions, so connected-component size, shape, and spatial continuity should be used to validate candidate objects.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

34. Thresholding Technique Comparison

Global thresholding uses one intensity threshold for the entire image and is fast, but it is sensitive to illumination variation. Adaptive thresholding calculates local thresholds and can handle nonuniform brightness, but may fragment smooth surfaces or amplify texture. Otsu thresholding selects a threshold from the image histogram and works best when foreground and background form separable distributions. A controlled lighting setup plus global or Otsu thresholding is a suitable baseline, with adaptive thresholding as a fallback.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

35. Edge Detection: Sobel, Canny, and Laplacian

Sobel estimates first-order intensity gradients and is computationally efficient for broad boundaries. Laplacian responds to second-order changes but is noise-sensitive. Canny combines smoothing, gradient estimation, non-maximum suppression, and hysteresis thresholds to produce thin connected edges. For battery outline extraction, Canny is generally the most complete candidate, while Sobel can be retained for fast local measurements.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

36. Morphological Processing

Erosion shrinks foreground regions, dilation expands them, opening removes small foreground noise, and closing fills small gaps. These operations can repair segmentation masks and stabilise object contours. The structuring element must be defined in physical units through the calibrated pixel scale. A kernel large enough to remove tiny specks could also erase a genuine 0.3 mm defect, so morphology should be restricted to object masks and not applied indiscriminately to defect evidence.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

37. Connected Components and Contour Analysis

Connected-component labelling groups adjacent foreground pixels into candidate objects. Contours provide perimeter, area, bounding rectangle, convex hull, and orientation. These measurements support form-factor classification and object tracking. Touching batteries may merge into one component; watershed or distance-transform splitting can be considered, but the system should also detect ambiguous overlaps and route them to a safe reject or manual review state.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

38. Region of Interest Extraction

After detecting a battery, the system crops a padded region of interest while preserving its original pixel scale. The ROI should include the full battery boundary and enough surrounding belt to estimate local illumination and background. Separate ROIs can be generated for the casing, end caps, label, and pouch perimeter. ROI coordinates must be transformed consistently after undistortion and geometric calibration.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

39. Form-Factor Classification Overview

The form-factor task distinguishes cylindrical 18650, cylindrical 21700, and prismatic pouch cells. Useful features include aspect ratio, contour compactness, circular end geometry, parallel side edges, width, length, and surface texture. Classification should combine metric measurements with shape descriptors rather than relying on one threshold, particularly when objects rotate or overlap.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

40. 18650 and 21700 Dimensional Analysis

The common nominal cylindrical dimensions are approximately 18 mm diameter by 65 mm length for 18650 and 21 mm diameter by 70 mm length for 21700. These dimensions provide a physical basis for classification, but real products and measurement tolerances vary. The image should be calibrated in millimetres, and classification boundaries should include a tolerance band to avoid false certainty near nominal limits.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

41. Prismatic Pouch Cell Identification

Pouch cells typically have a flattened rectangular envelope, flexible foil laminate, and sealed perimeter edges. Their contours are less rigid than cylindrical cans and can deform with swelling. The classifier should look for broad planar faces, elongated rectangular shape, corner geometry, and seal boundaries. A separate pouch-specific measurement path is needed because its safety condition depends on thickness and bulging, not only its outline.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

42. Shape Descriptor Comparison

Aspect ratio is fast and interpretable but can confuse differently sized objects when orientation changes. Circularity, defined as $4\pi A/P^2$, helps identify compact cylindrical end views but depends on contour quality. Hu moments provide scale- and rotation-invariant shape information but are more complex to interpret. A combined rule-based feature vector is a transparent first-stage classifier for the constrained CPU environment.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

43. Orientation Estimation

The principal axis of a battery can be estimated from the covariance matrix of its foreground coordinates or from a minimum-area rotated rectangle. This orientation allows the system to align the ROI before measuring length and width. End-cap features and text regions can then be searched in a consistent coordinate frame, reducing sensitivity to arbitrary placement on the belt.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

44. OCR Challenges on Laser-Etched Codes

The chemistry strings are small, high-value signals on glossy, reflective casings. Etching may produce low contrast, and glare can erase portions of characters. Codes such as LCO, LFP, NMC, and NiMH must be read without silently guessing. The OCR stage should return both the recognised string and a confidence score, and uncertain or conflicting results should be treated as unsafe for automatic acceptance.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

45. Text Region Detection

The text region can be found using local contrast, gradient density, connected components, and geometric constraints. The system should search likely label zones after estimating battery orientation. Morphological closing can join broken strokes, while filtering by component height and width can remove unrelated scratches. A separate text ROI reduces computation and helps keep casing defects from contaminating OCR.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

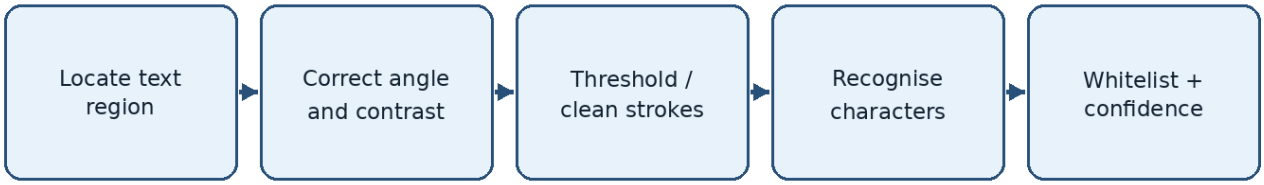
For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

46. OCR Preprocessing Methods

Contrast-limited adaptive histogram equalisation can improve local contrast without the extreme amplification associated with unrestricted histogram equalisation. Top-hat or black-hat transforms can emphasise bright or dark strokes relative to their surroundings. Adaptive thresholding may separate etched characters from uneven backgrounds. These methods should be evaluated on real reflective samples because the best polarity and threshold depend on illumination.

Figure: Chemistry-code OCR workflow



In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

47. Character Recognition and Validation

Recognition may use template matching, a classical OCR engine, or a compact trained classifier. Template matching is transparent and efficient when fonts and scale are stable, but it is brittle to wear and rotation. A trained recogniser handles more variation but needs representative labelled images. A chemistry-code whitelist and syntax checks can catch implausible outputs; validation must not turn an uncertain string into a confident but incorrect code.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

48. Chemistry Code Decision Logic

The recognised chemistry code is mapped to an approved routing category only after confidence and consistency checks. The scenario's critical example is lithium cells being mixed into a NiMH batch. A code that is unreadable, outside the whitelist, or inconsistent with other visual evidence should trigger a reject or quarantine decision. The system should preserve the original crop so an auditor can review the exact marking.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

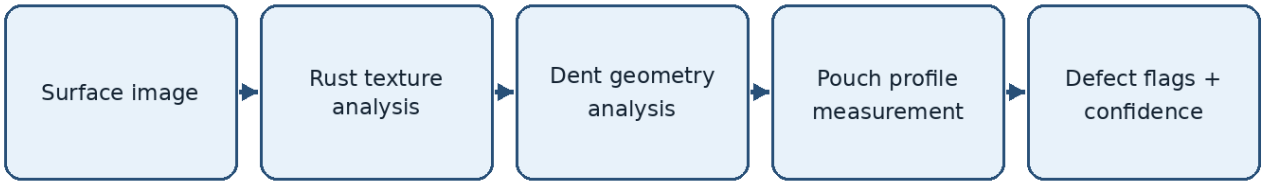
For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

49. Defect Detection Overview

The specified defects are surface rust spots, severe dents, and pouch-cell ballooning or swelling. These defects have different visual signatures: rust is often a low-contrast texture or colour change, dents alter local geometry and shading, and swelling changes the pouch profile. One universal threshold is unlikely to detect all three reliably; separate defect-specific analysis paths are recommended.

Figure: Defect-specific inspection branches



In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

50. Rust Spot Detection

The rust contrast is only 5–8 grey levels on an 8-bit scale, so acquisition quality is critical. A stable illumination field, higher bit depth, flat-field correction, and local contrast analysis help preserve the signal. Candidate rust regions can be identified using local intensity or colour deviation, texture statistics, and connected-component filtering. Validation should include clean brushed metal with similar texture to avoid false alarms.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

51. Low-Contrast Enhancement

Local contrast enhancement can make weak rust signals easier to separate from the casing. CLAHE operates on local tiles and clips histogram amplification to limit noise growth. Difference-of-Gaussians and local background subtraction can highlight small spots against slowly varying illumination. Parameters should be selected from labelled samples, with a fixed evaluation protocol to prevent overfitting to one lighting condition.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

52. Dent Detection Using Shape and Shading

A severe dent can produce a local deviation from the expected casing profile, a contour irregularity, or a characteristic highlight-shadow pair. Multi-directional lighting helps distinguish real surface deformation from a simple reflection. Shape comparison against an expected cylindrical or planar surface can identify abnormal regions. A single intensity threshold is not sufficient because glossy reflections can mimic dents.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

53. Pouch Swelling Measurement

The assignment requires swelling detection with dimensional precision of ±0.5 mm. A top-down image alone may not measure thickness accurately unless the pouch profile is visible or a calibrated geometric relationship exists. A side-view camera, structured light, or laser displacement sensor may be needed for direct thickness measurement. If only the overhead camera is allowed, the report should state the limitation and validate any top-surface bulge proxy against physical measurements.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

54. Measurement Uncertainty and Calibration

Measurement uncertainty includes pixel quantisation, lens distortion, camera mounting, focus, object height, edge localisation, belt vibration, and calibration-target error. A measurement reported to ±0.5 mm requires the combined uncertainty to remain within that tolerance across the full field. Calibration should be repeated after camera movement, lens adjustment, or maintenance, and reference artefacts should be inspected at scheduled intervals.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

55. Decision Fusion and Safety Logic

The final decision should combine form factor, chemistry, defect state, OCR confidence, object localisation confidence, and tracking status. A safe policy is fail-closed: unreadable chemistry, uncertain identity, severe defect, or missing association between image and conveyor track results in reject or quarantine. The system should distinguish a confirmed defect from an uncertain observation in its audit record.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

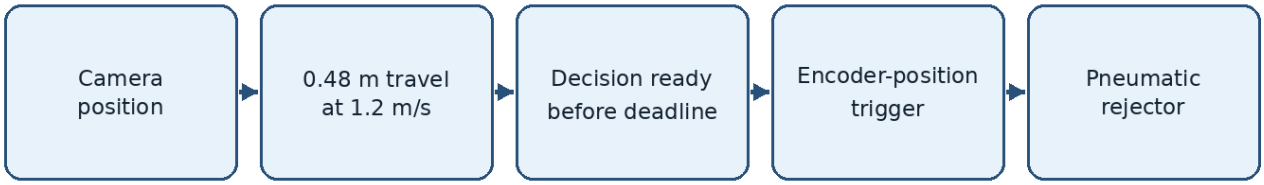
For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

56. Pneumatic Reject Timing

The piston is 0.48 m downstream from the camera. At a constant belt speed of 1.2 m/s, travel time is 0.48/1.2 = 0.4 s, or 400 ms. This exactly matches the stated decision deadline and leaves no nominal timing margin. The controller should use encoder position rather than a fixed delay, and the processing decision must be available well before the target reaches the reject station.

Figure: Conveyor tracking and reject timing



In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

57. Actuator Control and Interlocks

The reject output should be generated through an industrial digital I/O interface or PLC. The command must include object ID, reject reason, and scheduled encoder position. Interlocks should verify that the actuator is ready and that no emergency stop or maintenance mode is active. A missed actuation should be logged and should place the line into a safe state rather than silently allowing a potentially hazardous cell to pass.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

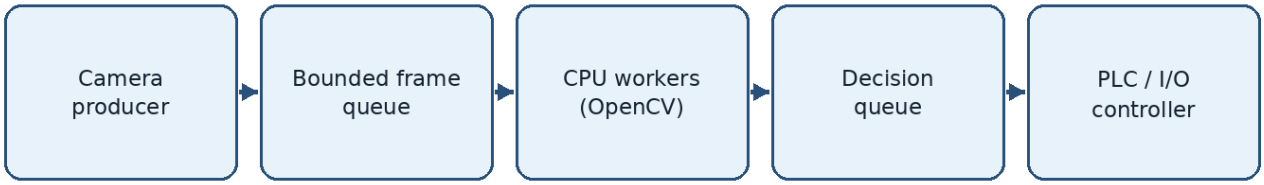
For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

58. Real-Time Software Architecture

A producer-consumer architecture separates camera acquisition from image processing and actuator scheduling. Acquisition writes frames into a bounded queue; processing workers consume frames and return decisions; a deterministic controller schedules the reject event. Queue depth, dropped frames, processing latency, and trigger jitter should be monitored. Overload must be detected explicitly, with a defined fail-safe response.

Figure: Real-time software architecture



In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

59. CPU and Memory Budget

The edge PC has four CPU cores and 16 GB RAM with no GPU. Full-resolution images should be processed efficiently using NumPy vectorisation and OpenCV primitives. Avoid unnecessary copies and repeated colour conversions. A bounded number of ROIs, reusable buffers, and controlled thread counts reduce memory pressure and prevent OpenCV's internal threading from oversubscribing the CPU.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

60. Python, OpenCV, and NumPy Implementation

Python provides rapid integration and audit-friendly code, OpenCV provides calibrated imaging, filtering, morphology, contours, and geometry, and NumPy supports efficient array operations. Critical real-time functions should avoid Python-level pixel loops. Profiling must use representative full-resolution frames and concurrent tasks. If latency is excessive, optimise ROI processing, downsample only noncritical stages, or move selected operations to compiled OpenCV routines.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

61. Image Compression and Archival Policy

The plant must store visual proof of every processed cell for five days on a 1.5 TB SSD. Lossless or lightly compressed formats preserve evidence, while JPEG reduces storage at the cost of artefacts that may affect tiny defects and OCR. A practical strategy is to retain a high-quality original or lossless crop for audit, plus a compact metadata record and optional annotated image. The selected policy must be tested against the storage budget.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

62. Storage Capacity Calculation

At 15 cells per second, the maximum daily count for continuous 24-hour operation is 1,296,000 cells. Five days gives 6,480,000 images. At 100 kB per stored image, the total is about 648 GB decimal; at 200 kB it is about 1.30 TB before filesystem overhead and metadata. Therefore average compressed image size, operating hours, and retention enforcement must be measured and budgeted.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

63. Data Traceability and Audit Records

Each record should contain a unique cell ID, capture timestamp, encoder position, image filename, form-factor result, chemistry result, defect flags, confidence values, final decision, and actuator acknowledgement. The record must link to the unmodified evidence image. Clock synchronisation and access controls protect audit integrity. Logs should be append-oriented and include software version and calibration status.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

64. Hardware Budget Allocation

The total hardware budget is ₹1,25,000. A feasible allocation reserves the largest share for the global-shutter camera and lens, then lighting and enclosure, followed by trigger/encoder hardware, industrial I/O, mounting, and storage. Actual prices vary by supplier and specification, so procurement should obtain quotations and preserve contingency. A low-cost camera that cannot resolve the 0.3 mm defect is a false economy.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

65. Candidate Design Comparison

Three practical acquisition configurations can be compared: a low-cost rolling-shutter colour camera with continuous light; a global-shutter monochrome camera with strobed diffuse light; and a higher-resolution global-shutter camera with multi-angle controlled lighting. The first is cheapest but vulnerable to motion distortion and glare. The second is a balanced baseline. The third offers the strongest defect visibility but must be checked against the budget and CPU throughput.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

66. Recommended Imaging Front-End

The recommended baseline is a global-shutter monochrome industrial camera with approximately 4096 pixels across the 320 mm field, a low-distortion lens, short exposure, and synchronised high-intensity LED lighting. A diffuse source supports text and general inspection; multiple low-angle sources support defects. The choice is provisional until sample testing confirms the 0.3 mm defect and OCR stroke requirements at the actual working distance.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

67. Verification Dataset and Ground Truth

Testing must include representative 18650, 21700, and pouch cells; all listed chemistry codes; clean and defective surfaces; multiple rotations; partial occlusions; and the full range of ambient conditions. Ground truth should be assigned by trained inspectors and, for dimensions, by calibrated physical instruments. The dataset should include borderline examples rather than only ideal images.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.

68. Performance Metrics

Classification should be evaluated using a confusion matrix, per-class precision, recall, and F1 score. Defect detection should report sensitivity, specificity, false rejects, and missed-defect rate. OCR should report exact-string accuracy and character error rate. Measurement should report bias, repeatability, and absolute error. Average latency alone is insufficient; the 95th and 99th percentile processing times must also meet the line deadline.

In the GreenCycle sorting line, this topic must be evaluated against the fixed overhead camera height of 400 mm, the 320 mm × 240 mm field of view, and the maximum conveyor speed of 1.2 m/s. The system cannot assume that the factory environment is optically stable: ambient illumination changes from 1,500 lux to 45,000 lux and overhead lighting introduces 100 Hz flicker. Consequently, the selected method should be tested with controlled illumination and representative production samples rather than tuned on a single laboratory image.

The engineering decision should be supported by measurable evidence. Relevant observations include spatial resolution in pixels per millimetre, contrast-to-noise ratio, edge localisation error, false-positive rate, missed-defect rate, processing latency, and repeatability across object orientation. Results should be reported separately for cylindrical 18650 cells, cylindrical 21700 cells, and prismatic pouch cells because their geometry and surface response differ. A method that performs well on one form factor must not be assumed to generalise to the others.

Implementation should preserve the original acquired image and record the processing parameters used for each decision. Python, OpenCV, and NumPy are sufficient for many classical vision operations when they are applied to bounded regions of interest and implemented with vectorised or compiled routines. The four-core CPU and absence of a GPU make predictable runtime important. Profiling should include worst-case images, overlapping objects, glare, low-contrast rust, and difficult OCR examples.

For acceptance, the method must contribute to a complete safety case: chemistry recognition must not silently convert uncertainty into a valid class; defects must be detected with a miss rate compatible with the stated 1.5% limit; and the final decision must be available early enough to schedule the rejector. If a required measurement cannot be proven from the overhead image alone, the limitation must be documented and a suitable additional sensor or quarantine rule proposed.

ENGINEERING TAKEAWAY: Select the method only after measuring accuracy, repeatability, and latency under the stated factory constraints. Preserve uncertainty and route unsafe or ambiguous observations to reject or quarantine.